MODULE 28 · DOMAIN 3 - COST-OPTIMIZED ARCHITECTURE DESIGN

Content Delivery and Data Transfer Costs

Serve from the edge, shield the origin, keep traffic private and in its own Availability Zone, and pick the network path that charges least per gigabyte.

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Every gigabyte has a **path**, and every hop on the path may charge. Shorten the path, or move it onto free hops.

Where gigabytes cost money

PATH	CHARGE
Within one AZ, private IP	Free
Between AZs in a Region	Charged in each direction
Between Regions	Charged per GB, higher
To the internet	Charged; lower through CloudFront
AWS origin to CloudFront	Free
NAT gateway, TGW, interface endpoint	Per-GB processing on top

ON THE EXAM

Processing charges stack: a byte through a NAT gateway and across AZs pays twice.

CloudFront as a cost tool

- **Cache hits** never reach the origin: less compute and less egress.
- **Origin fetches** from AWS origins carry no data transfer charge.
- **Cache policies**: forward only the headers and query strings that matter.
- **Origin groups** fail over to a secondary origin on errors.
- **Price classes** limit edge locations to cut cost where reach allows.

> ON THE EXAM
>
> A low cache hit ratio is usually a cache key that includes things that do not change the response.

Longer TTLs, or Origin Shield?

TUNE CACHING

Fewer misses at each edge

- Longer TTLs where content allows
- Narrower cache keys
- Versioned file names
- Free to change

ORIGIN SHIELD

One extra cache layer

- A Regional cache in front of the origin
- Edges' misses collapse into one fetch
- Charged per request through it
- Best with global audiences or several CDNs

THE TIEBREAKER

Caching already tuned, origin still flooded by misses from many locations: Origin Shield.

Private paths to AWS services

PATH	HOURLY	PER GB
Gateway endpoint (S3, DynamoDB)	None	None
Interface endpoint (PrivateLink)	Per AZ	Low processing charge
NAT gateway	Per gateway	Processing charge, higher
Regional NAT gateway	Spans AZs automatically	Processing charge

ON THE EXAM

Heavy S3 or DynamoDB traffic through a NAT gateway: add free gateway endpoints.

Availability Zone-aware design

- **Keep chatty tiers in the same AZ**: zonal affinity for calls and caches.
- **NAT and endpoints in every AZ** used, so traffic stays local.
- **Load balancer cross-zone** settings affect where requests go.
- **Replicas read locally**: read from the replica in your own AZ.
- **Never trade resilience for this**: stay multi-AZ, just route locally.

> ON THE EXAM
>
> Cut inter-AZ charges by routing within each zone, not by collapsing into one zone.

Connecting VPCs: cost per gigabyte

OPTION	PER-GB PROCESSING	FITS
VPC peering	None; data transfer only	A few high-volume pairs
Transit Gateway	Per GB, per attachment hop	Many VPCs, central routing
PrivateLink	Per GB, lower	Expose one service, overlapping CIDRs
Cloud WAN	Per GB plus core network	Global, policy-based networks

ON THE EXAM

One pair of VPCs moving huge volumes through a transit gateway: add a peering connection for that pair.

CloudFront flat-rate plans

- **Monthly tiers**: Free, Pro, Business and Premium, since November 2025.
- **Bundles** delivery, WAF and DDoS protection, Route 53 DNS, logs and edge compute.
- **No overage charges**, even during traffic spikes or attacks.
- **Pay-as-you-go** remains; choose per distribution.

> ON THE EXAM
>
> Predictable website cost with security included, no surprise bills: a flat-rate plan.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Global downloads served from one Region	CloudFront with S3 origin
Origin flooded despite good TTLs	Origin Shield
S3 or DynamoDB traffic through NAT	Gateway endpoints
High inter-AZ charges between tiers	Zonal affinity, stay multi-AZ
One heavy VPC pair on a transit gateway	Peer that pair
Predictable cost, fear of attack bills	CloudFront flat-rate plan

ON THE EXAM

Trace the path, find the charged hop, and move the traffic to a free or cheaper one.

> Meridian Freight's customers worldwide download route-planning map packs of 200 MB each directly from an S3 bucket in eu-west-1. Downloads are slow outside Europe.
>
> Data transfer out is now the largest item on the bill.

What should the solutions architect recommend?

A. Enable S3 Transfer Acceleration on the bucket

B. Serve the map packs through a CloudFront distribution with the bucket as an origin using origin access control

C. Replicate the bucket to six Regions with Cross-Region Replication

D. Move the bucket to the S3 Express One Zone storage class

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Transfer Acceleration

Adds a fee per GB, and every download still comes from the bucket.

B CloudFront with OAC

Cached at the edge, free origin fetches, cheaper delivery.

C Replicate to six Regions

Six copies of storage plus replication transfer.

D Express One Zone

For low-latency access in one AZ, not global delivery.

THE SIGNAL

Popular objects downloaded worldwide: cache them at the edge.

> Meridian Freight streams live depot camera feeds to viewers worldwide through CloudFront, from a single origin in us-east-1. Cache settings are already tuned.
>
> Requests from many edge locations still reach the origin for the same segments, and the origin is struggling.

What should the solutions architect add?

A. A second origin in another Region

B. AWS Global Accelerator in front of the origin

C. CloudFront Origin Shield in us-east-1

D. Lambda@Edge to rewrite requests

◆ PAUSE AND DECIDE

What should the solutions architect add?

A Second origin

Doubles origin infrastructure instead of reducing fetches.

B Global Accelerator

Speeds the network path; caches nothing.

C Origin Shield

Collapses edge misses into one fetch per object.

D Lambda@Edge

Adds compute, not a cache layer.

THE SIGNAL

Many edges missing on the same objects, caching already tuned: Origin Shield.

> Meridian Freight's analytics instances in private subnets read 40 TB a month from S3 and write heavily to DynamoDB. All this traffic leaves through NAT gateways.
>
> NAT gateway processing charges have become significant.

What should the solutions architect do?

A Add S3 and DynamoDB gateway endpoints to the private subnets' route tables

B Add interface endpoints for S3 and DynamoDB

C Move the instances to public subnets with public IP addresses

D Replace the NAT gateways with larger NAT instances

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Gateway endpoints

No hourly or per-GB charge; traffic bypasses NAT.

B Interface endpoints

Private, but still charged hourly and per GB.

C Public subnets

Weakens security to save money.

D NAT instances

More to operate; traffic still detours.

THE SIGNAL

S3 and DynamoDB from inside a VPC: gateway endpoints, which are free.

> Meridian Freight's pricing service runs in three AZs and calls a cache cluster many times per request. Cost reports show large inter-AZ data transfer charges between the two tiers.
>
> The service must remain resilient to the loss of an AZ.

What should the solutions architect recommend?

A. Move both tiers into a single AZ

B. Add a transit gateway between the tiers

C. Use larger cache nodes to reduce the number of calls

D. Route each service instance to cache nodes in its own AZ, keeping capacity in all three AZs

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Single AZ

Removes the charge and the resilience together.

B Transit gateway

Adds a per-GB processing charge.

C Larger nodes

The call count comes from the application, not node size.

D Zonal affinity

Calls stay within each zone; all zones still serve.

THE SIGNAL

Inter-AZ charges with a resilience requirement: route within each zone.

> Meridian Freight connects 60 VPCs through a transit gateway. One analytics VPC pulls about 300 TB a month from a data VPC, and this pair accounts for most of the transit gateway's processing charges.
>
> The CIDR ranges do not overlap, and everything else should stay on the transit gateway.

What should the solutions architect recommend?

A. Move all 60 VPCs to VPC peering

B. Add a VPC peering connection between the analytics and data VPCs, with more specific routes for that pair

C. Add a second transit gateway for the analytics traffic

D. Expose the data VPC through AWS PrivateLink

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Peer all 60

Thousands of connections to manage, for one heavy pair.

B Peer the heavy pair

No processing charge for that traffic; the rest stays central.

C Second transit gateway

Still charges per GB processed.

D PrivateLink

Lower per-GB charge, but not zero, and built for one service.

THE SIGNAL

One heavy pair, non-overlapping CIDRs: peer it, and keep the hub for everything else.

> Meridian Freight's marketing website runs on CloudFront and S3. The small marketing team wants a predictable monthly bill that includes WAF and DDoS protection.
>
> Last year a traffic spike during a bot attack produced an unexpected bill.

What should the solutions architect recommend?

A Subscribe to AWS Shield Advanced

B Keep pay-as-you-go pricing and add an AWS Budgets alert

C Move the distribution to a CloudFront flat-rate pricing plan

D Put AWS Global Accelerator in front of CloudFront

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Shield Advanced**

Strong protection, but a large monthly fee for a marketing site.

B **Budgets alert**

Reports the surprise; does not prevent it.

C **Flat-rate plan**

Fixed monthly price with WAF and DDoS included, no overages.

D **Global Accelerator**

Adds cost and does nothing for billing predictability.

THE SIGNAL

Predictable bill, security included, no spike charges: CloudFront flat-rate plan.

What to carry out of this module

- **Trace the path**: distance charges plus per-GB processing hops.
- **CloudFront** for popular content; **Origin Shield** when edges flood the origin.
- **Gateway endpoints** for S3 and DynamoDB; never pay NAT for them.
- **Zonal affinity**, not single-AZ, to cut inter-AZ charges.
- **Peer heavy pairs** beside a transit gateway; flat-rate plans for predictability.

> ON THE EXAM
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> Move bytes onto free or cheaper paths without giving up resilience.